

RF-Based 12-BIT SIGNAL TRANSMITTER And RECEIVER

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The circuit presented here converts 12-bit parallel digital data to serial data and then transmits it through RF transmitter module. Serial data is received at RF receiver module in serial format. Received signal is then converted to parallel data through a microcontroller (MCU), same as original data.

This circuit can be modified for use as a remote water-level indicator, remote controller for a robot, remote LCD system, or remote digital data logger and monitoring system.

Circuit and working

The circuit diagram of the RF-based 12-bit signal transmitter and receiver is shown in Fig. 1. It is built around 5V voltage regulator 7805 (IC1), Atmega8A MCU (IC2), 433MHz RF transmitter (TX1) and receiver (RX1)

modules, dip switch (DIP1) and a few other components.

For conversion of 12-bit parallel data to serial data, and then of serial data to parallel data, ATmega8A MCU is used. It has inbuilt, separate serial data-out and serial data-in pins (USART). 12-bit parallel data is available digitally on the 12 pins of the MCU.

The circuit can be wired and used in two ways (modes) with slight modification, namely, transmission (TX) mode and receiving (RX) mode.

TX mode. If shorting jumper SJ2 is selected for high (5V), MCU is in transmission mode. The circuit can convert 12-bit parallel data (D0 to D11) available at connector CON3 to serial data and transmit it through its RF transmitter module TX1. RF receiver RX1 module is not required in this case.

RX mode. If SJ2 is selected for low (ground), MCU is in receiving mode. The circuit can receive serial data through RX1 and convert it to 12-bit parallel data (D0 to D11), which is available at CON3.

The same program is used for both transmitter and receiver sections. But for multiple functions, settings are done using SJ1 and SJ2. Option (OPT) selection is done using SJ1 to select encoding, or auto/manual transmission. Options are read by the MCU whenever it is powered up or reset by pressing reset (RST) switch S2. Program selection (PS) is done using SJ2.

SJ1 works differently with the combination of SJ2 selection as listed in the table (next page).

In TX mode, if SJ1 is high, continuous transmission happens at regular intervals. This is initially set in the

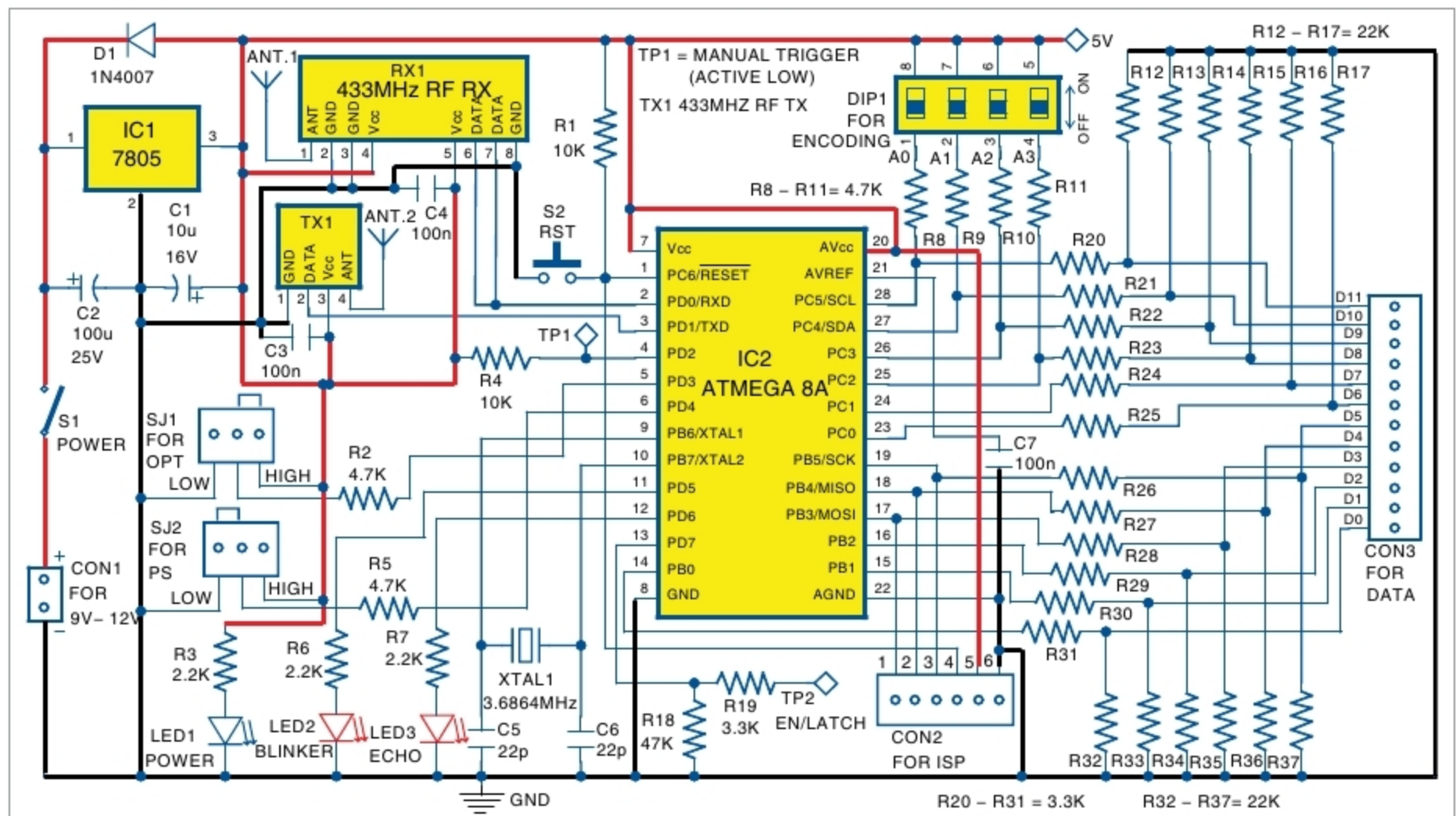


Fig. 1: Circuit diagram of 12-bit signal transmitter and receiver